

Beam-Spline Mappers in the Mok FSI Case

Theory, Current Implementation, Verification, and Stability Evidence

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1 Scope and canonical setup

This note is the single maintained technical record for `FSI_mok_Test_Case`. It explains the current implementation of `BeamSplineMapper` and `BeamSplineMapper.WithRotationalRecovery`, the analytical benchmark design, the conservative inverse maps, and the retained nonlinear FSI evidence. Historical parameter sweeps and pre-fix output are not part of the current evidence set.

The physical model follows the official Kratos Mok validation case. The two-dimensional wall occupies

$$x \in [0.4965, 0.5015] \text{ m}, \quad y \in [0, 0.25] \text{ m},$$

and its beam abstraction is placed on the centroidal line $x = 0.499$ m. The formal linear-thickness scale is $s = 50$. It is not a numerical tuning parameter: structure and fluid load are scaled together. The beam properties are

$$\begin{aligned} A &= 0.25, & I_{22} &= 2.083333333333336 \times 10^{-2}, \\ I_{33} &= 5.208333333333334 \times 10^{-7}, & J &= 2.083333333333334 \times 10^{-6}. \end{aligned}$$

Fluid `REACTION` is multiplied by +50 before transfer, and `swap_sign` remains active. The retained FSI gate uses

$$\Delta t = 0.05, \quad \alpha = 0.03, \quad N_{\text{coupling}} = 60.$$

2 Plain BeamSplineMapper

2.1 Forward map

The plain mapper is a one-dimensional beam-spline construction. A surface node is projected onto the reference beam chain. Axial translation and torsion are interpolated on the local beam coordinate; transverse centerline translations use spline systems whose derivative data are supplied by the beam rotations. A fixed affine augmentation $\{1, \xi\}$ is used, with

$$\xi = \frac{x - X_c}{a} \in [-1, 1],$$

where X_c is the reference-chain midpoint and $a = L/2$. Derivative rows are scaled by $1/a$. This normalization removes dependence on coordinate translation and avoids the dimensional imbalance of an unscaled beam coordinate.

The destination displacement has the generic form

$$u_d = F_{\text{plain}}(q_b).$$

Its section-offset contribution uses the recovered/interpolated section rotation. Because a finite rotation matrix is used in that contribution, F_{plain} is nonlinear in the rotation degrees of freedom even though the spline solves themselves are linear.

2.2 Conservative inverse

The load map is defined by discrete virtual work:

$$\delta u_d = J_{\text{plain}}(q_b) \delta q_b, \quad Q_b = J_{\text{plain}}(q_b)^\top f_d.$$

The bending contribution therefore cannot be distributed only to the two nodes of the projected segment. Its evaluation functional is first accumulated over all destination forces and is then propagated to the whole beam support through the transpose spline solve. The normalized derivative scaling used in the forward map is also used in this adjoint.

Resolved correctness issue. The earlier inverse path used only a two-node force/lever-arm distribution and was not the transpose of the global spline interpolation. It also omitted part of the section offset induced by bending rotations. The repaired map uses the exact forward tangent and its transpose. The current four directional virtual-work errors are

$$9.88 \times 10^{-10}, \quad 4.52 \times 10^{-10}, \quad 1.41 \times 10^{-10}, \quad 2.40 \times 10^{-9}.$$

All pass the 10^{-7} tangent/adjoint acceptance criterion.

3 BeamSplineMapper with rotational recovery

3.1 Paper-faithful small-rotation field

Ahrem, Beckert, and Wendland reconstruct one vector displacement field $s : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. At structural sites X_i , the generalized interpolation conditions are

$$s(X_i) = g_i, \quad \nabla \times s(X_i) = 2h_i.$$

The second equation follows from infinitesimal kinematics,

$$h = \frac{1}{2} \nabla \times u,$$

and the paper explicitly restricts the interpretation to small angles. With three value and three curl functionals per point, the saddle system is

$$\begin{bmatrix} A_{\Phi, X} & P_X \\ P_X^\top & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} g \\ 2h \\ 0 \end{bmatrix}.$$

In `rotation_recovery_mode = small`, the surface displacement is evaluated directly:

$$u_s = s(X_s).$$

For fixed support this map is linear, $u_s = Hq_b$, so the conservative load map is $Q_b = H^\top f_s$.

3.2 Polynomial levels

The public integer setting `polynomial_level` has the following meaning:

- 0 : automatic rank/conditioning fallback,
- 1 : paper full-3D affine space, 12 vector modes,
- 2 : line-basic space, 9 modes,
- 3 : line-enriched space, 12 modes,
- 4 : centered/scaled Legendre high-order space, 30 modes.

Level 4 replaces dimensional monomials by Legendre polynomials in $\xi \in [-1, 1]$ without reducing the polynomial span. Level 1 can be non-unisolvent on a perfectly collinear value/curl support and is

rejected when its functional matrix is rank deficient. Level 0 selects the highest full-rank level below the conditioning limit; it selected level 4 with a polynomial condition estimate of approximately 1.86×10^2 in the analytical beam test.

Legacy string-valued `polynomial_basis` values are translated with a deprecation warning. Supplying both interfaces is an error.

3.3 Kernel, coordinates, and regularization

For the Kratos Gaussian convention

$$\phi(r) = \exp\left[-\frac{1}{2}(r/h)^2\right],$$

the paper radius d in $\exp(-r^2/d^2)$ satisfies $d = \sqrt{2}h$. All recovery coordinates and their value/curl evaluation functionals use the same centered reference scale. The analytic Gaussian Hessian at $r = 0$,

$$\nabla\nabla\phi(0) = -\frac{1}{h^2}I,$$

is retained in the curl-curl self block. The previous generic zero-distance branch incorrectly replaced this nonzero limit by zero.

Regularization is added only to the kernel/interpolation block. The polynomial constraint block remains exactly zero, as required by the saddle formulation. A rank defect in P is never hidden by regularizing this zero block. The production recovery setting is

$$h = 0.50, \quad \epsilon = 10^{-8}, \quad \text{polynomial_level} = 4.$$

3.4 Finite rigid-section hybrid

The selectable `rotation_recovery_mode = finite` is deliberately labelled an Ahrem-rigid-section hybrid, not a finite-rotation theorem from the Ahrem paper. Let X_c be the reference projection on the beam and $r_0 = X_s - X_c$. The Ahrem solve provides

$$u_c = s(X_c), \quad \theta_c = \frac{1}{2}\nabla \times s(X_c).$$

The surface displacement is then

$$u_s = u_c + [\exp([\theta_c]_\times) - I] r_0.$$

The reference offset is never formed from current node coordinates. Rodrigues' formula and small-angle series are used for the exponential map. No rotation-vector recovery from R is performed, avoiding an additional π -branch operation.

This model enforces a rigid section and exactly reproduces finite rigid section rotations. It is not the same physical model as direct evaluation of the three-dimensional Ahrem displacement field: the latter can represent a general field and does not impose a rigid cross-section.

The finite map is nonlinear. Its conservative inverse is the current-state tangent transpose,

$$Q_b = J(q_b)^\top f_s, \quad J = \frac{\partial u_s}{\partial q_b}.$$

For $\theta = \phi n$, the offset tangent uses the analytical SO(3) right Jacobian

$$J_r(\theta) = I - \frac{1 - \cos \phi}{\phi^2}[\theta]_\times + \frac{\phi - \sin \phi}{\phi^3}[\theta]_\times^2.$$

The resulting nodal moment is work-conjugate to the selected rotation-vector parameter. At finite rotation it is not generally identical to a naive unmodified lever-arm moment. The state used by the inverse is cached during the corresponding forward map; inverse-before-forward has an explicit `DISPLACEMENT/ROTATION` fallback and warning.

4 Common geometric and algebraic corrections

The following defects were resolved in both mapper implementations.

1. Mapper search no longer writes reference coordinates into the beam nodes' current coordinates. Projection may use reference geometry, but it has no observable side effect on the structural state.
2. When an approximate projection coordinate is clamped, its shape functions, projection point, distance, and section offset are rebuilt from that same clamped coordinate.
3. The local frame is constructed from the Cartesian axis least aligned with the beam tangent, followed by orthogonal projection and a right-handed cross product. Negative axes, near-axis directions, and nearly collinear segments no longer enter discontinuous equality branches.
4. Reference-chain midpoint and half-length scaling are used consistently in values, derivatives, curls, moment functionals, and inverse functionals.
5. Explicit inverse matrices are not stored. Forward systems are solved, and inverse functionals are aggregated per beam chain before solving $A^T z = b$.
6. `swap_sign` is applied only at the physical load accumulation; it is not mixed into the mathematical transpose.

5 Analytical benchmark design

Mapper accuracy is measured only against prescribed analytical fields, never against another mapper. For surface-node values,

$$e_{\text{relL2}} = \sqrt{\frac{\sum_i \|u_i^{\text{map}} - u_i^{\text{exact}}\|^2}{\sum_i \|u_i^{\text{exact}}\|^2}}.$$

Exact-span tests with zero regularization use a practical solver-level target of 10^{-10} , relaxed to 10^{-9} when the assembled saddle solve and coordinate transforms make 10^{-10} marginal. Fields that centerline values and curl cannot uniquely identify are reported as *non-identifiable diagnostics*, not exact-reproduction failures.

5.1 Forward verification

The core suite contains rigid translation and infinitesimal rotation, axial motion, bending, torsion, curl-compatible fields, polynomial-level reproduction, nonuniform spacing, translated/scaled coordinates, negative axes, and near-collinear geometry. Finite-mode checks prescribe exact rigid rotations of $1^\circ, 30^\circ, 90^\circ, 170^\circ$. Every test also checks that mapper construction leaves current coordinates unchanged.

The current rigid-rotation results are:

	1°	30°	90°	170°
BeamSplineMapper	5.81×10^{-6}	5.22×10^{-3}	4.51×10^{-2}	8.83×10^{-2}
Recovery small	9.18×10^{-4}	2.89×10^{-2}	1.13×10^{-1}	3.13×10^{-1}
Recovery finite	3.98×10^{-15}	4.47×10^{-16}	5.71×10^{-16}	1.18×10^{-15}

These results quantify kinematic compatibility; they do not establish FSI stability.

5.2 Backward/adjoint verification

For a prescribed state q , perturbation p , and surface force f , the centered finite-difference directional derivative is compared with the inverse-map work:

$$\frac{f^\top [F(q + \varepsilon p) - F(q - \varepsilon p)]}{2\varepsilon} \quad \text{versus} \quad p^\top J(q)^\top f.$$

The production Jacobian is analytical; finite differences are used only for verification. Bending about both transverse axes, torsion, and a mixed field are included. All current plain, recovery-small, and recovery-finite cases pass. Their worst relative discrepancies are respectively

$$2.40 \times 10^{-9}, \quad 2.24 \times 10^{-9}, \quad 2.57 \times 10^{-9}.$$

6 FSI benchmark design

The FSI runner supports NearestNeighbor, BeamMapper_CoRotation, BeamSplineMapper, and both recovery modes, together with scale, time step, relaxation, coupling-iteration limit, kernel radius, regularization, polynomial level, structural strategy, and target end time. Each trial:

1. snapshots every modified JSON, MDPA, and material file;
2. synchronizes section and fluid-load scaling;
3. writes point history, CFD/CSM VTK, full log, settings, conclusion, and summary JSON into a unique directory;
4. monitors nonfinite output, coupling failure, and repeated structural nonconvergence;
5. stops invalid runs and verifies restoration by file hashes.

The stability gate is $0.5 \rightarrow 4.5 \rightarrow 10 \rightarrow 25$ s. Failure of one gate prevents an unsupported longer run. FSI trials do not use NearestNeighbor as an accuracy reference; the analytical suite supplies accuracy, while FSI supplies nonlinear stability and failure-mechanism evidence.

7 Retained post-fix FSI results

All retained trials use $s = 50$, $\Delta t = 0.05$, $\alpha = 0.03$, 60 coupling iterations, synchronized scaling, and `swap_sign`.

Mapper/mode	Target	Last valid	Failure
BeamSplineMapper, normalized	4.5 s	3.05 s	structural nonconvergence, step 62 at 3.10 s
Recovery, level 4, small	4.5 s	4.20 s	structural nonconvergence, step 85 at 4.25 s
Recovery, level 4, finite	4.5 s	3.05 s	structural nonconvergence, step 62 at 3.10 s

All three trial summaries report successful restoration. No mapper reaches the 4.5 s gate, so no post-fix run was promoted to 10 or 25 s. The finite recovery and normalized plain histories are almost identical at their limit: their point-A u_x values at $t = 3.05$ s are 0.01611986 and 0.01612157 m. Recovery-small reproduces the earlier step-85 limit after coordinate normalization and the Legendre polynomial correction. Therefore the old high-order monomial conditioning was not the root cause of that event.

The analytical checks exclude coordinate mutation, a discontinuous projection definition, missing Gaussian self-Hessian, a non-adjoint load map, and an incorrect finite $\text{SO}(3)$ tangent as current

explanations. FSI fields remain finite up to the last accepted step, after which the structural nonlinear solve repeatedly fails. The evidence presently classifies the terminal event primarily as a structural/coupled-trajectory nonconvergence, not as a mapper singularity or mapper-produced NaN. A structure-only replay of steps 62 and 85 is the next mechanism-isolating experiment.

8 Reproducibility and retained records

The canonical scripts are under `Scripts/`:

- `run_fsi_benchmark.py`: parameterized, restoring FSI runner;
- `validate_beam_spline_forward.py` and `validate_beam_spline_adjoint.py`;
- `validate_recovery_forward.py` and `validate_recovery_adjoint.py`;
- the conditioning, Newton-history, VTK-history, and load-consistency analyzers.

Current analytical results are in `TestCase_Output/MapperVerification`. Current nonlinear FSI evidence is restricted to the three $t = 4.5$ gate directories under `TestCase_Output/StabilityTrials_s50`. Every numerical conclusion above can be reconstructed from their summaries and logs.

9 Remaining theoretical decisions

1. Confirm whether the desired recovery mapper is the paper-faithful, general three-dimensional Ahrem displacement field or a rigid-section finite-rotation beam surface model. They are different nonlinear FSI models.
2. Confirm the precise semantics of the structural element’s `ROTATION`: total exponential-coordinate vector, incremental rotation, or another composed parameter. The finite hybrid requires a compatible total-state interpretation.
3. Confirm whether rigid cross-sections are a desired physical constraint. If section warping is relevant, the finite hybrid is too restrictive.
4. If a fully geometrically exact recovery is required, consider separate interpolation of translation and orientation on $SO(3)$, with a consistent tangent transpose, rather than assigning finite physical meaning to $\frac{1}{2}\nabla \times s$.